COL-341 HW-1

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2020CS10840

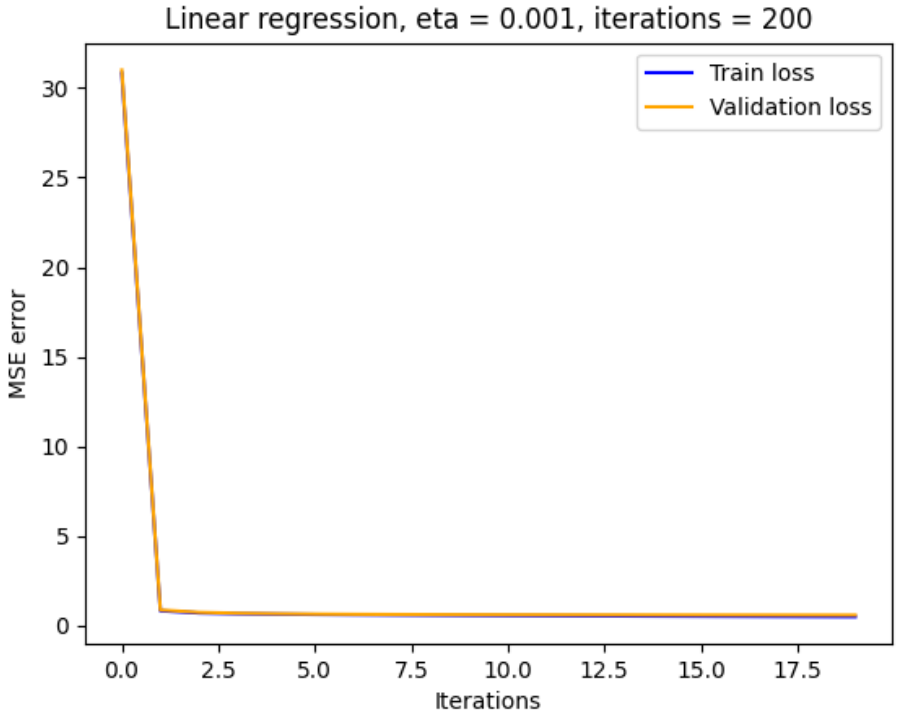
February 2023

Basic implementation

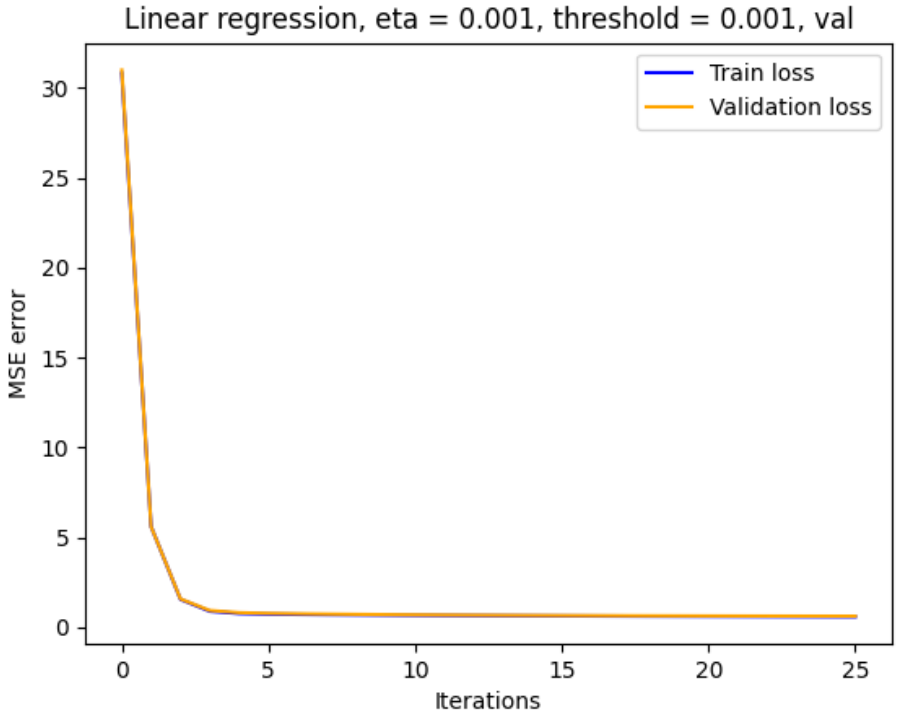
Analysing on different learning rates

NOTE - The line for training and testing loss were coming quite close to each other and it might be difficult to view the blue line

- $\eta = 0.1$
  - In this case the step size is too large and the error on training as well as validation data increases. The parameters diverged from the optimal value i.e they overshot the optimal values and kept on moving away from them
- $\eta = 0.01$
  - In this case the step size is too large and the error on training as well as validation data increases. The parameters diverged from the optimal value i.e they overshot the optimal values and kept on moving away from them
- $\eta = 0.001$
  - For 200 iterations MSE and MAE on training data were 0.488 and 0.551 respectively and on validation data their corresponding values were 0.578 and 0.579. From the graph it can be seen that in about 20 iterations the rate of decrease of training loss decrease and when the iterations become much larger overfitting might also start

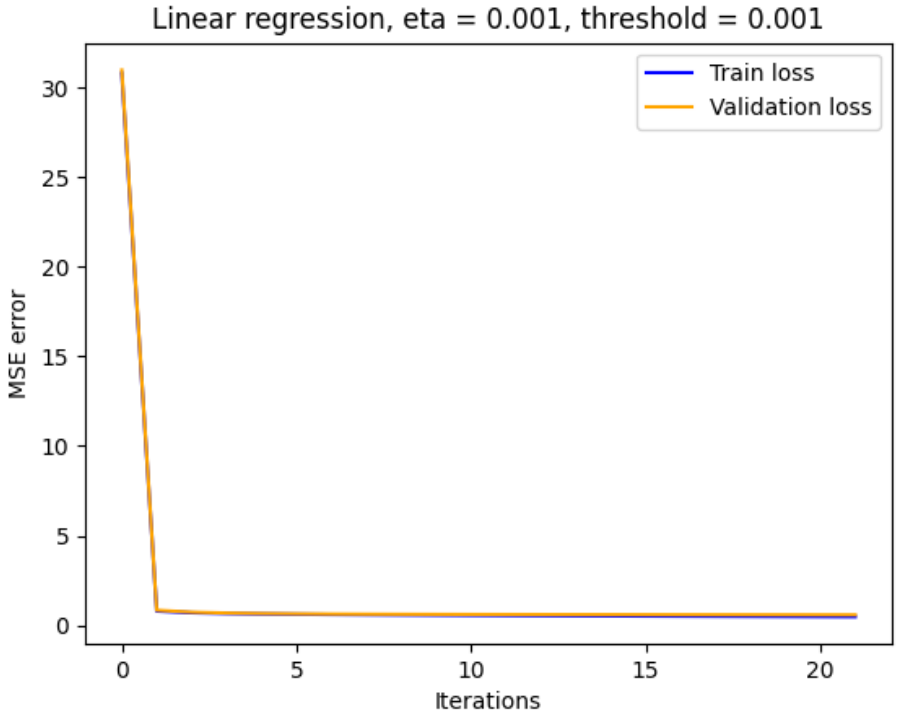


– Threshold of $<= 0.001$ in relative MSE loss on validation data was chosen as a criteria for stopping. The algorithm took 78 iterations. MSE and MAE on training data were 0.574 and 0.593 respectively and on validation data their corresponding values were 0.609 and 0.604.



– Threshold of $<= 0.001$ in relative MSE loss on training data was chosen as a criteria for stopping. The algorithm took 237 iterations. MSE and MAE on training data were 0.463 and 0.541 respectively and on validation data their corresponding values were 0.574 and 0.576. From the graph it can be seen that in about 20 iterations the rate of decrease of

training loss decrease and when the iterations become much larger overfitting might also start



Ridge Regression

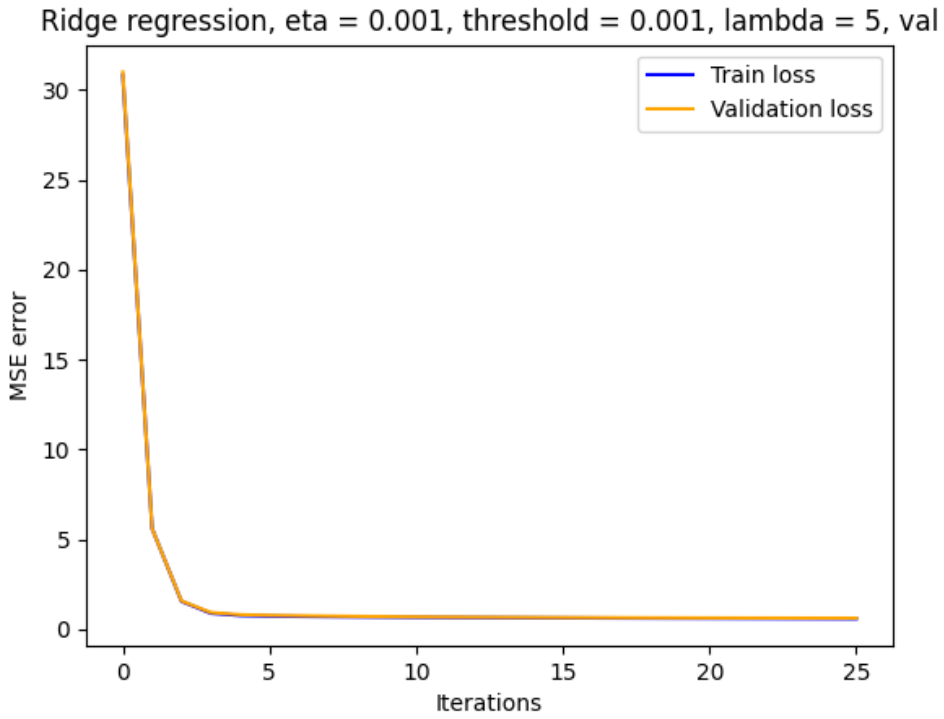
NOTE - The line for training and testing loss were coming quite close to each other and it might be difficult to view the blue line

Loss function for ridge regression E(θ)= 1/N($||\mathbf{y}-\mathbf{X}\theta||^2+\lambda\theta^T\theta$) = $1/2N(\theta^T\mathbf{X}^T\mathbf{X}\theta-2\theta^T\mathbf{X}^T\mathbf{y}+\lambda\theta^T\theta)$

Taking the gradient wrt θ we get = 2/N($\mathbf{X}^T\mathbf{X}\theta - \mathbf{X}^T\mathbf{y} + \lambda\theta$)

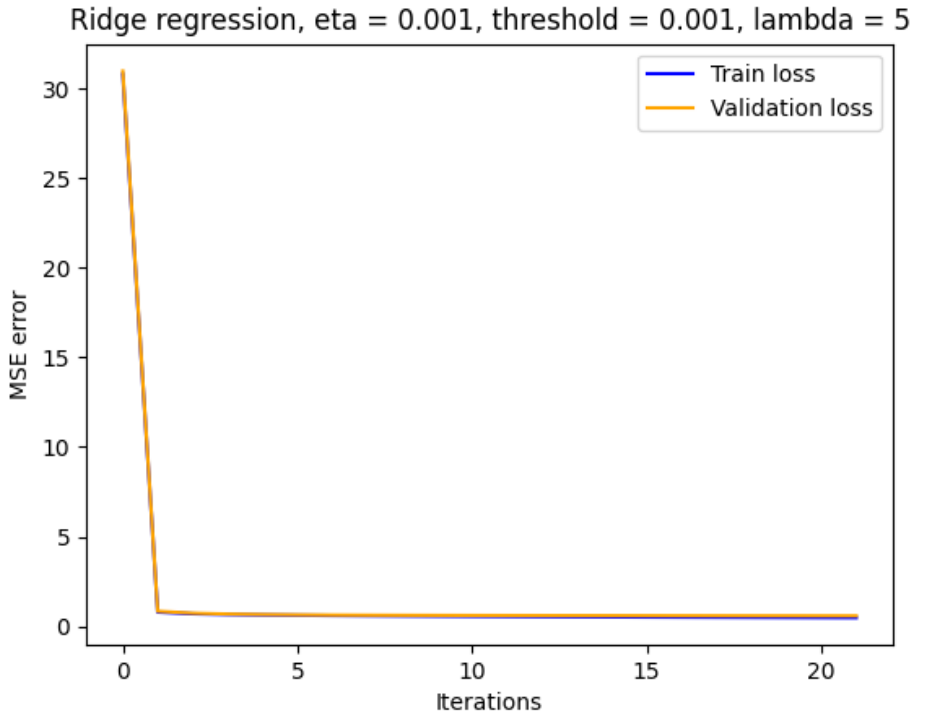
Hence the parameter update equation would be $\theta = \theta - 2/N(\mathbf{X}^T\mathbf{X}\theta - \mathbf{X}^T\mathbf{y} + \lambda\theta)$

- $\lambda = 5$. Taking $\eta = 0.001$, Relative threshold on validation data = 0.001, the algorithm converges in 78 iterations. The MSE and MAE loss on training data are 0.574 and 0.594 respectively and on validation data their corresponding values are 0.608 and 0.604.



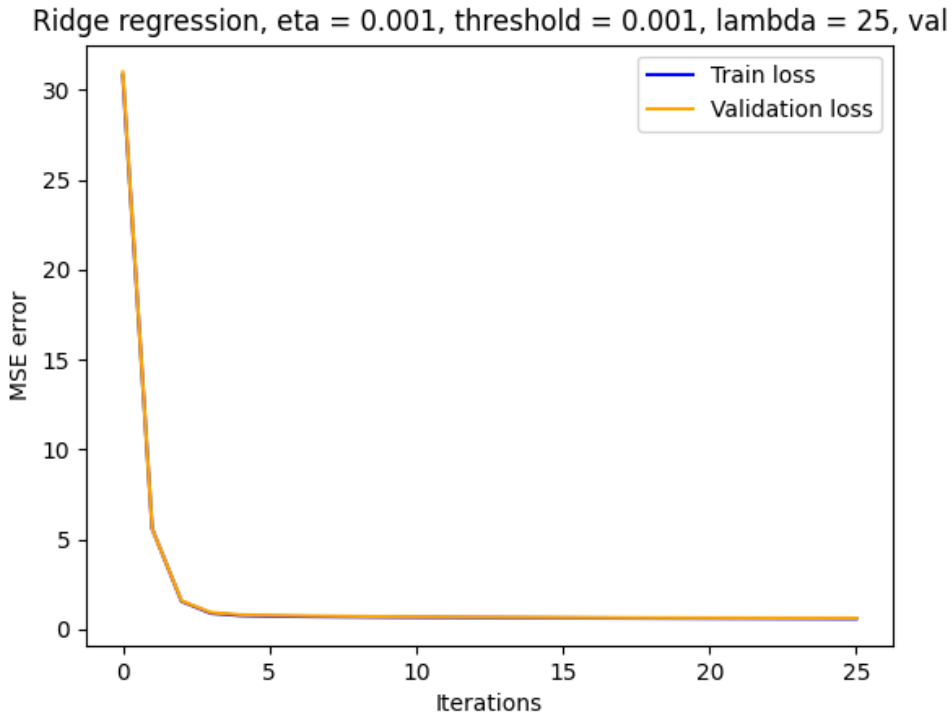
- $\lambda = 5$. Taking $\eta = 0.001$, Relative threshold on train data = 0.001, the algorithm converges in 235 iterations. The MSE and MAE loss on training data are 0.467 and 0.543 respectively and on validation data their corresponding values are 0.573 and 0.577.

  As compared to linear regression the mse loss value is slightly more on training data (0.467 vs 0.463) indicating higher bias. But the values were slightly lower as in validation data (0.573 vs 0.574)



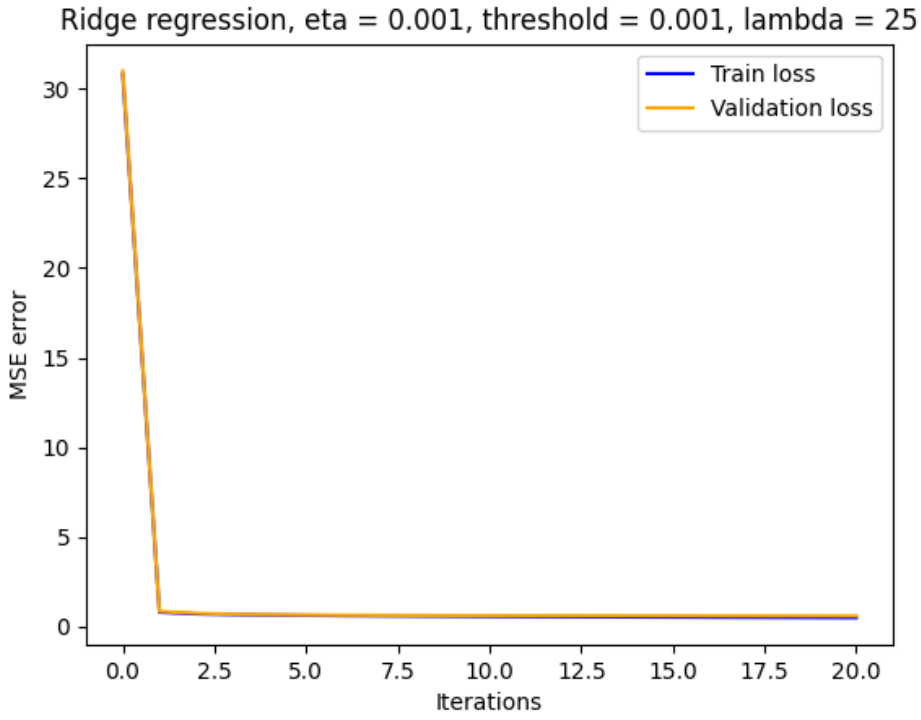
- $\lambda = 25$. Taking $\eta = 0.001$, Relative threshold on validation data = 0.001, the algorithm converges in 78 iterations. The MSE and MAE loss on training data are 0.575 and 0.594

respectively and on validation data their corresponding values are 0.607 and 0.604.



- $\lambda = 25$. Taking $\eta = 0.001$, Relative threshold on training data $= 0.001$, the algorithm converges in 231 iterations. The MSE and MAE loss on training data are 0.467 and 0.544 respectively and on validation data their corresponding values are 0.573 and 0.577.

  As compared to linear regression the loss value is slightly more on training data (0.467 vs 0.463) indicating higher bias. But the values were slightly lower as in validation data (0.573 vs 0.574)



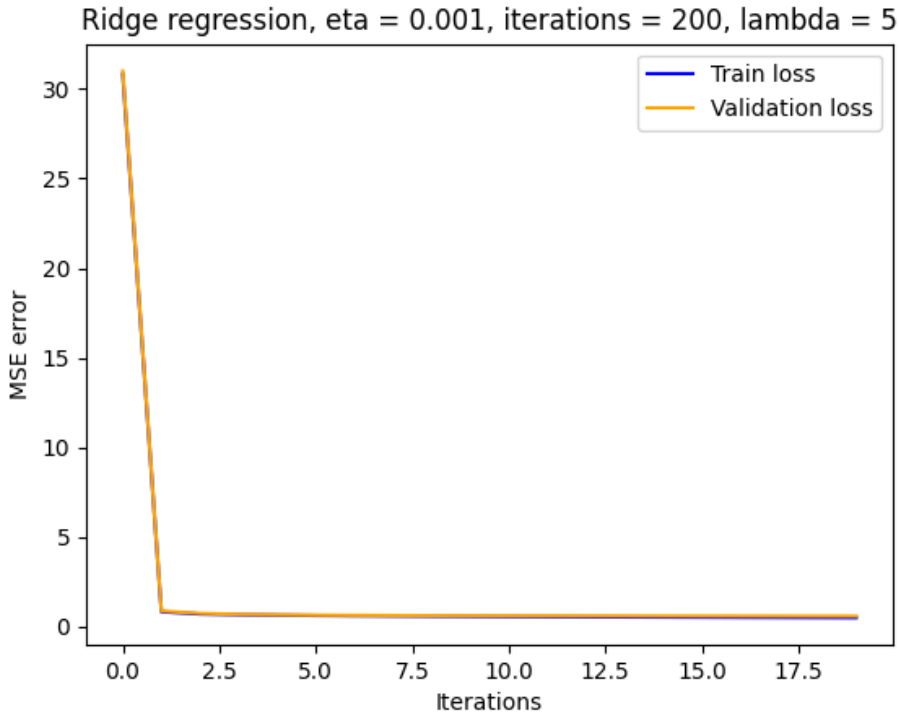
- This shows that ridge regression has higher bias than linear regression but lower variance i.e lin-

ear regression generalizes well and has better accuracy in validation data. Using ridge regression helps in over-fitting as we are associating an additional cost with each of the parameter. The higher the lambda the less likely we are to overfit as the parameter values are kept much smaller (because they have cost associated with them). This in some sense decreases the degrees of freedom for the model and allows better generalization

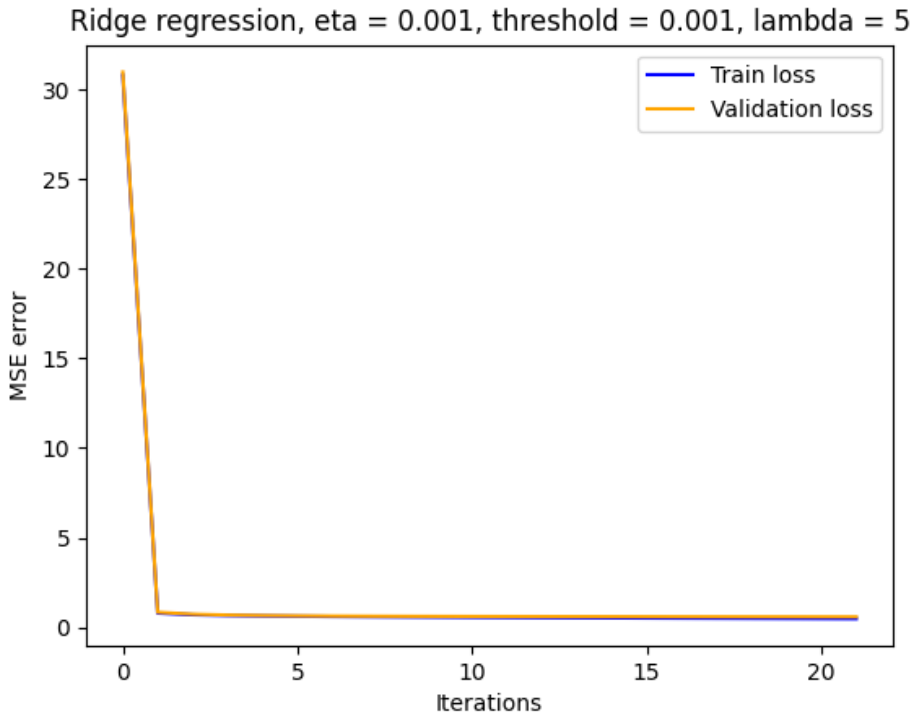
- Ridge regression also takes fewer iterations to achieve the same relative error threshold for the same learning rate

Similarly below are the observations for ridge regression with fixed number of iterations

- $\lambda = 5$. Taking iterations $=$ 200 and relative error threshold $=$ 0.001 The MSE and MAE loss on training data are 0.4824 and 0.551 respectively and on validation data their corresponding values are 0.578 and 0.579.



- $\lambda = 25$. Taking iterations $=$ 200 and relative error threshold $=$ 0.001, the algorithm converges in 235 iterations. The MSE and MAE loss on training data are 0.483 and 0.552 respectively and on validation data their corresponding values are 0.578 and 0.579.



Using Scikit-Learn Library

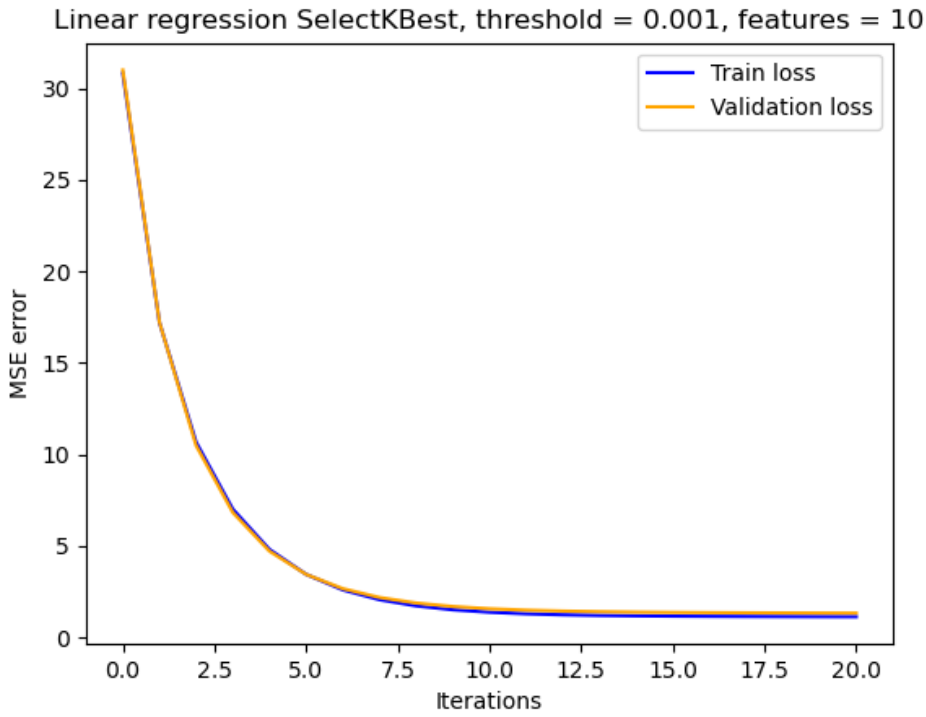
The linear regression model of sklearn library performs much better in the training data as compared to linear regression with the MSE coming in order of 10^{-29} and MAE coming in order of 10^{-15}. The data overfitted so much because the size of dataset was small 144 elements however the number of features were very high 2048.

However on validation data set the sklearn library performs poorly with a MSE of 1.024 as compared to 0.566(self-implementation) and MAE of 0.83 as compared to 0.559(self implementation)

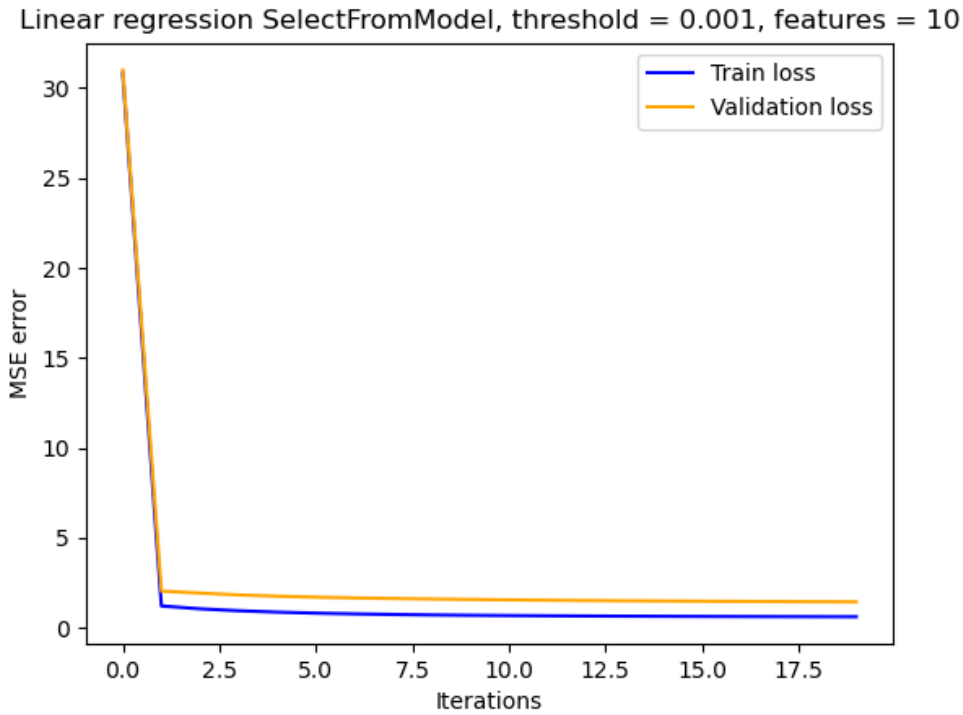
This clearly shows that sklearn's linear regression model overfits the data and has lower bias but very much higher variance and generalizes poorly as compared to previous implementation of linear and ridge regression

Feature Selection

- SelectKBest (K=10). The algorithm took 104 iterations till convergence on the top 10 features. The MSE and MAE on training data were 1.15 and 0.894. The MSE and MAE on validation data were 1.357 and 0.980



- SelectFromModel (Maximum parameters = 10). The algorithm took 149 iterations till convergence on the top 10 features. The MSE AND MAE on training data were 0.624 and 0.620. The MSE and MAE on validation data were 1.472 and 0.839



- The losses were higher than those obtained when taking all the 2048 features.
- However since the data has much fewer dimensions we can take much lower relative error thresholds in the same number of iterations
- For a given error threshold the number of iterations are also very few around 70 as compared

to 300. This could be much more scalable method in very large dimension dataset

Classification

Update equation

Consider a class $1 <= r <= 8$.

Taking $1 + \Sigma_{r=1}^{r=8}(e^{\theta_r x}) = \alpha$, the loss function function when described with respect to class r of θ_r is given by . $(-1/N)\Sigma_{i=1}^{N}((I(y^i = r)log((e^{\theta_r x}))/\alpha) + (I(y^i \neq r)log(1/\alpha))$

Taking the gradient of loss with respect to θ_r

$(1/N)\Sigma_{i=1}^{N}(-I(y^i = r) + I(y^i = r)((e^{\theta_r^T x})/(\alpha)) + I(y^i \neq r)((e^{\theta_r x}))/(\alpha))x$

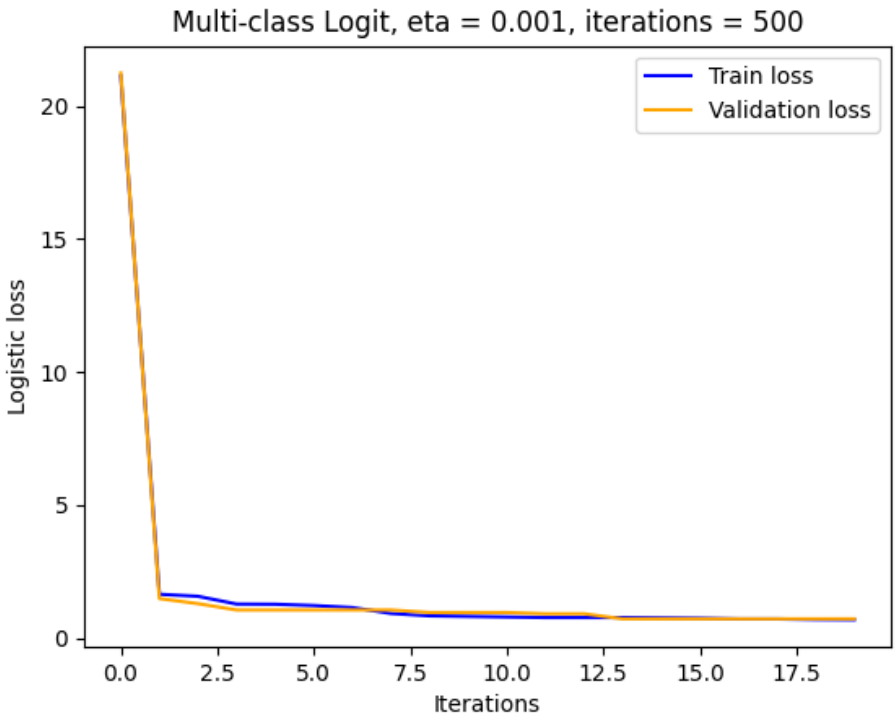
Hence the update equation is given by $\theta_r = \theta_r - \eta * ((1/N)\Sigma_{i=1}^{N}(-I(y^i = r) + I(y^i = r)((e^{\theta_r^T x})/(\alpha)) + I(y^i \neq r)((e^{\theta_r x}))/(\alpha))x)$

Accuracy

An accuracy of 0.57 on validation data was reached with 400 iterations and learning rate of 0.001. On increasing the number of iterations too much the model was clearly overfitting with accuracy on training data becoming 0.97 but accuracy on validation data becoming 0.47 (5000 iterations)

MSE loss

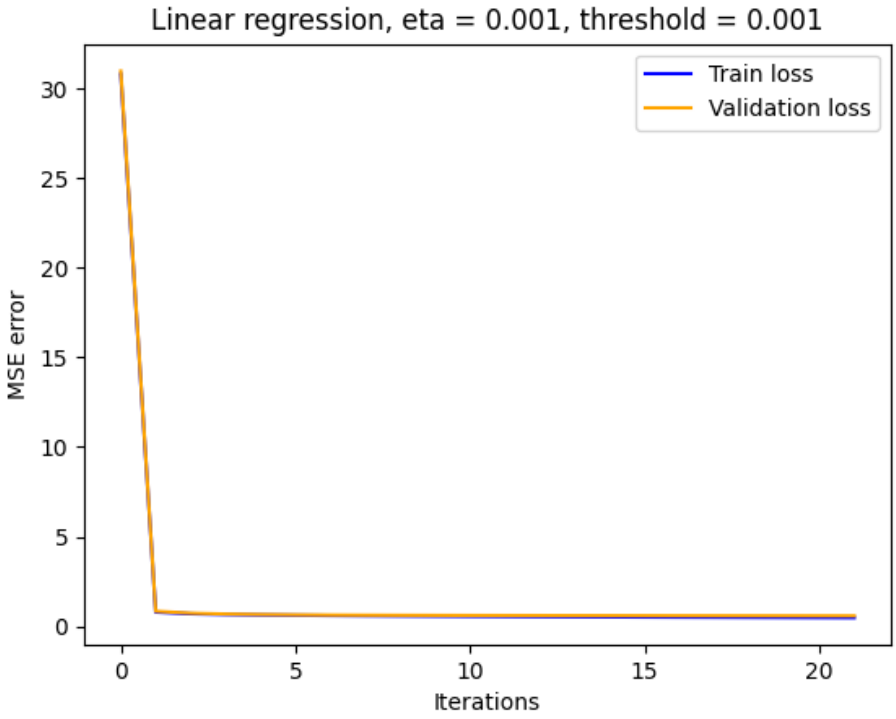
The smallest MSE loss on validation data was about 0.71 which was worse than that of linear regression (0.57)



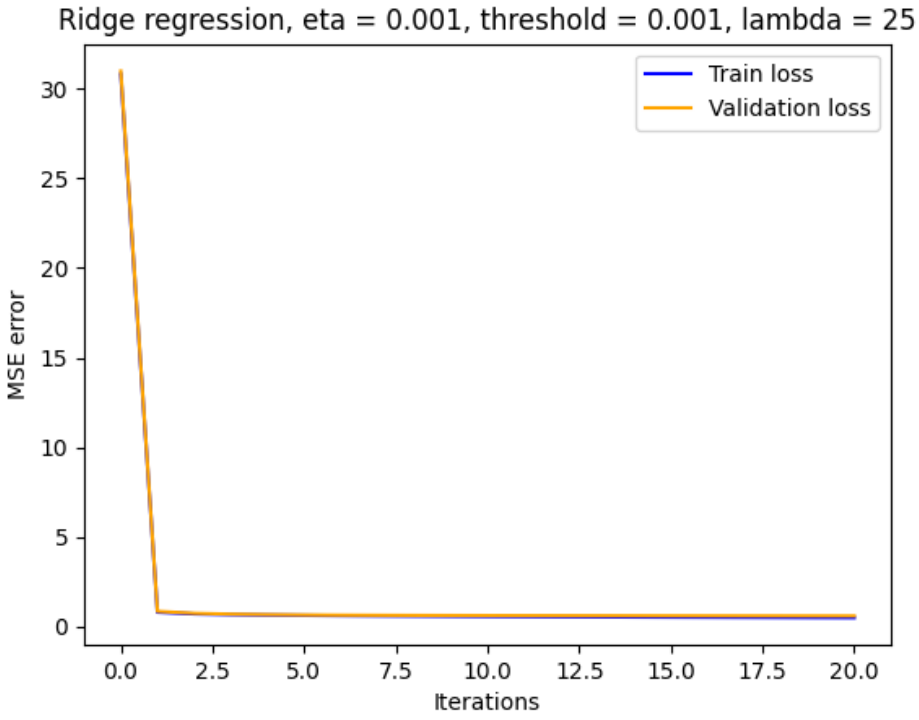
Visualization

Comparison of different models

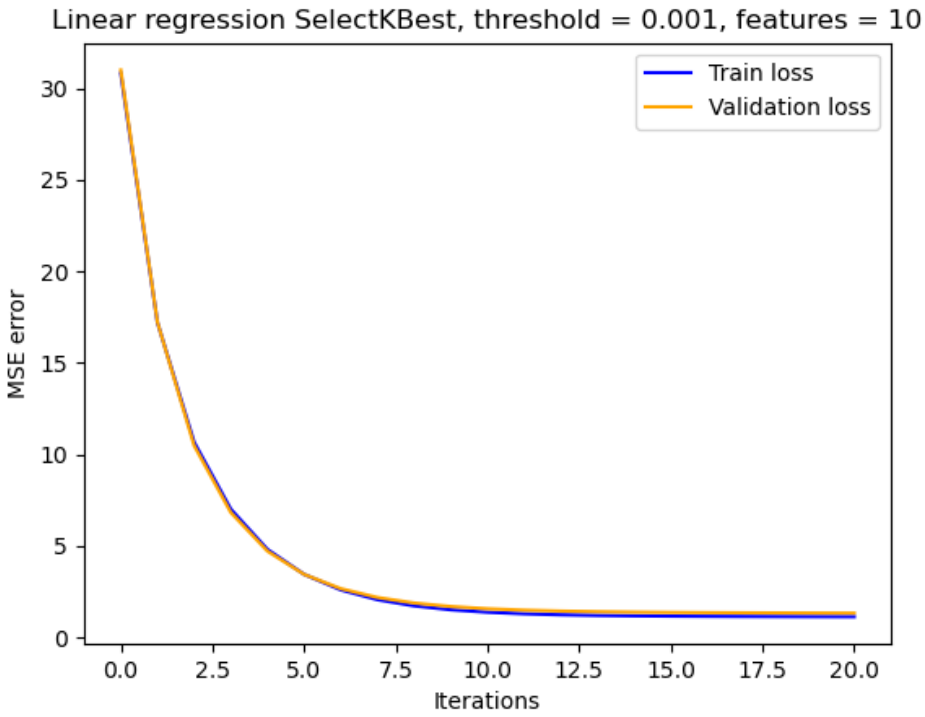
Linear regression with eta = 0.001 and all the features



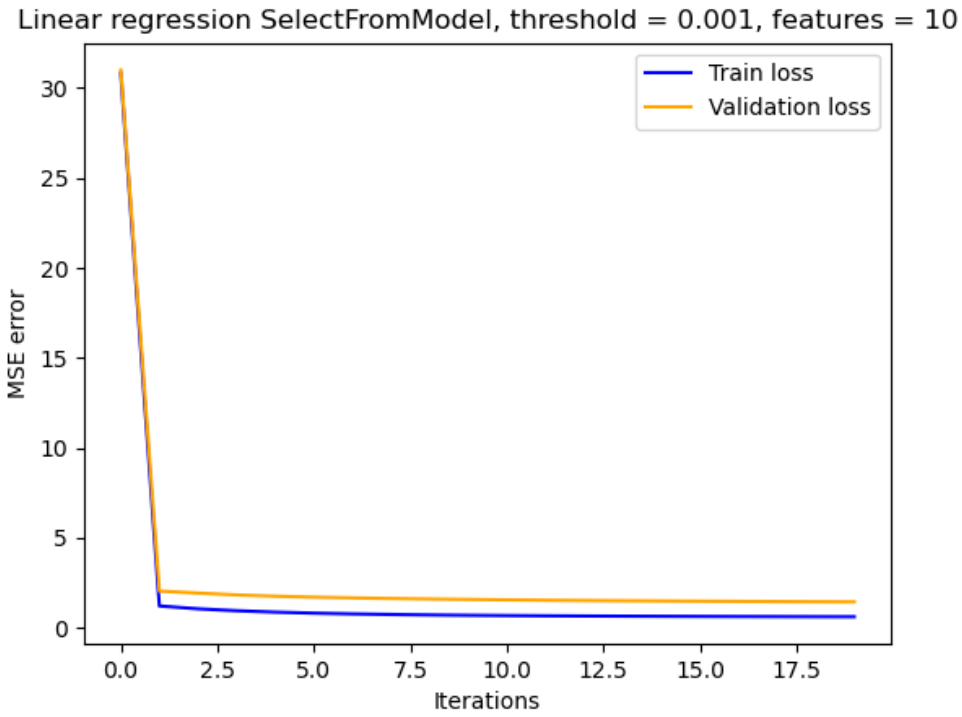
Ridge regression with eta = 0.001, lambda = 25 and all the features



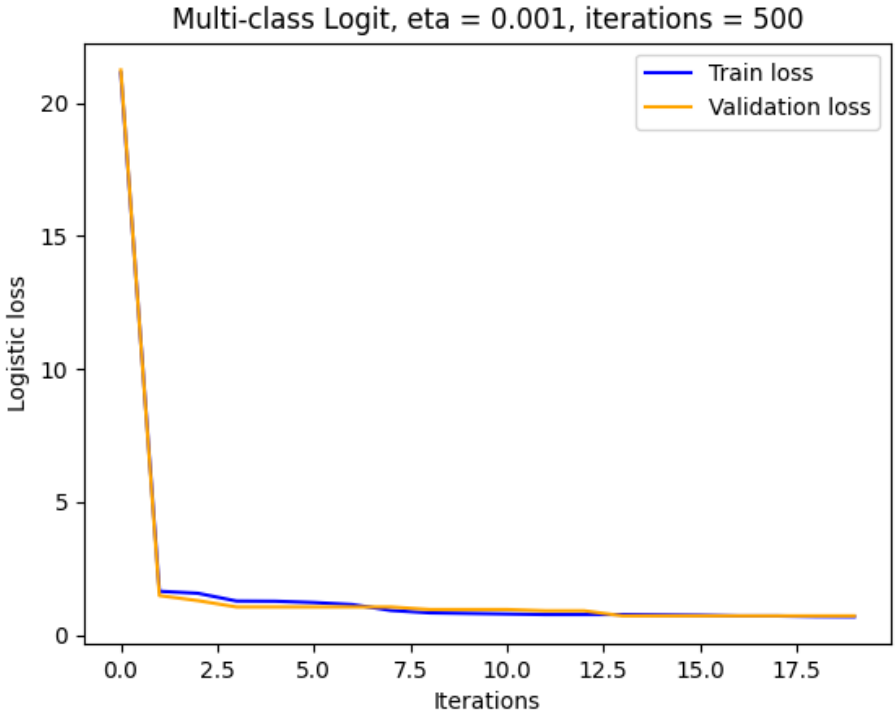
SelectKBest, 10 features, eta = 0.001



SelectFromModel, 10 features, eta = 0.001



Logistic regression with 500 iterations eta = 0.001

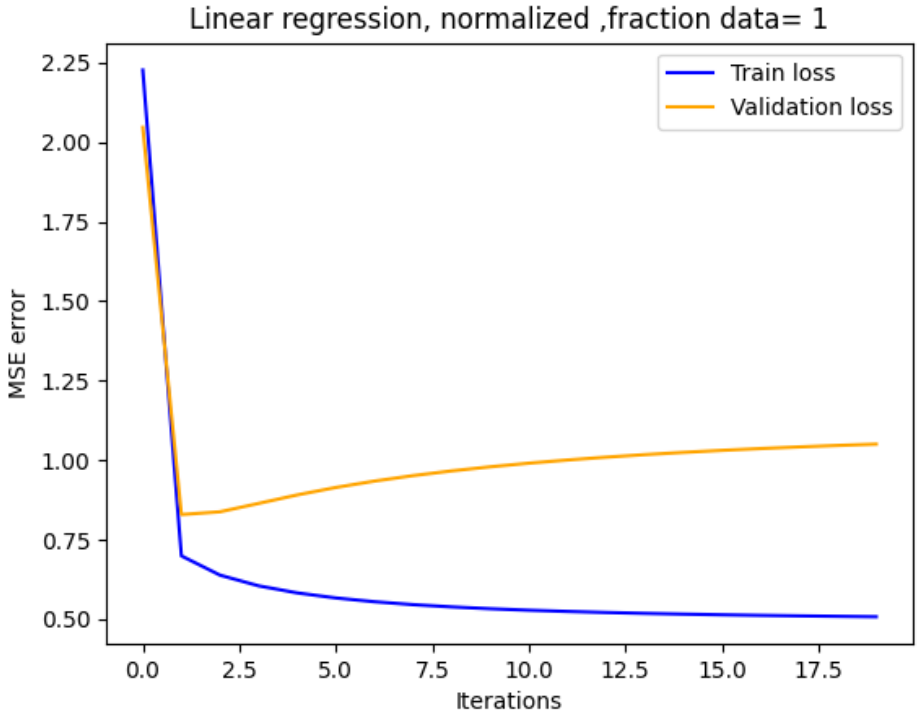


Observations- The error decreases smoothly for the model trained on top 10 features of SelectKBest. Overall the MSE loss on validation data was lowest for ridge regression with all the features. The model on top 10 features was the most scalable and took the least number of iterations. It could be also be improved by increasing error threshold. Logistic regression had a loss of 0.7 of validation data (between that of linear regression with all features and with less number of features)

Normalization

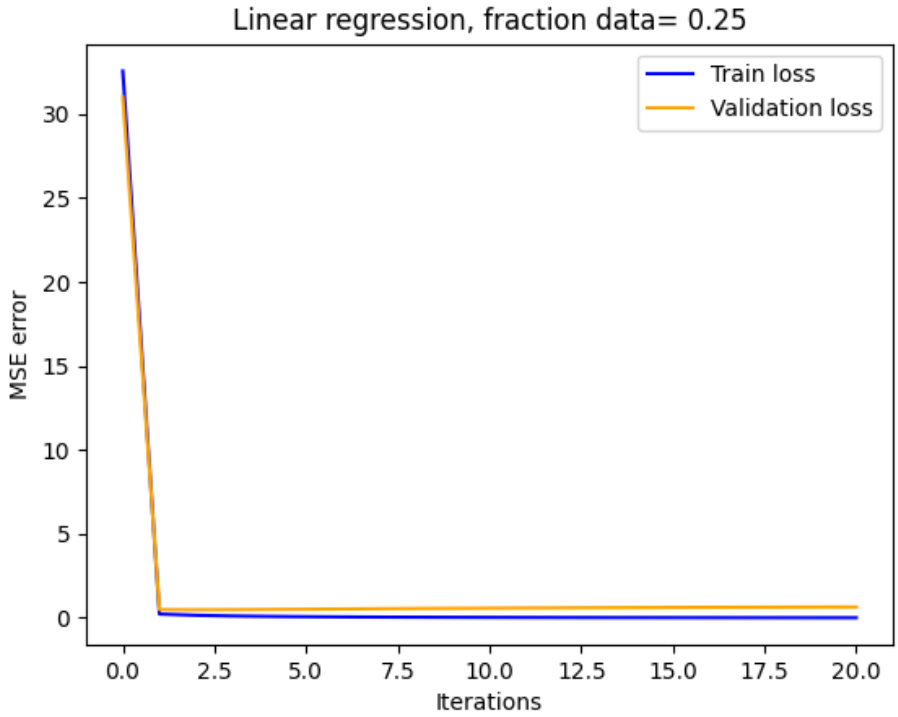
Observation - In case of normalization the model was easily over-fitting the training data.The MSE and MAE on training data were 0.50 and 0.61 and on test data were 1.054 and 0.845. The model performed poorly as compared to the use of unnormalized data.

Normalized data with $\eta = 0.001$ and 1000 iterations

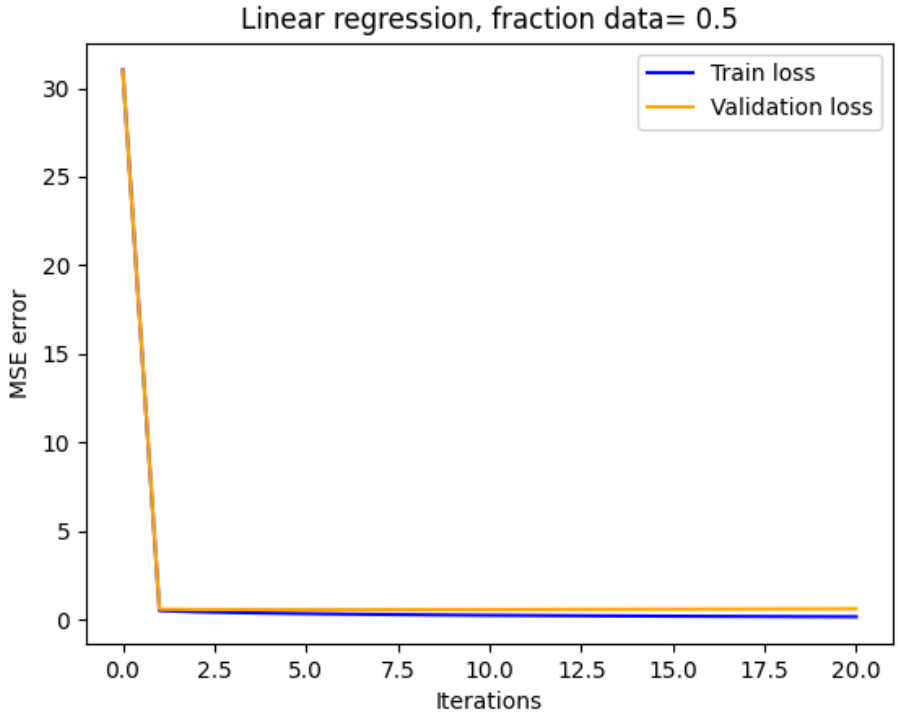


Plotting loss for different fractions of training data

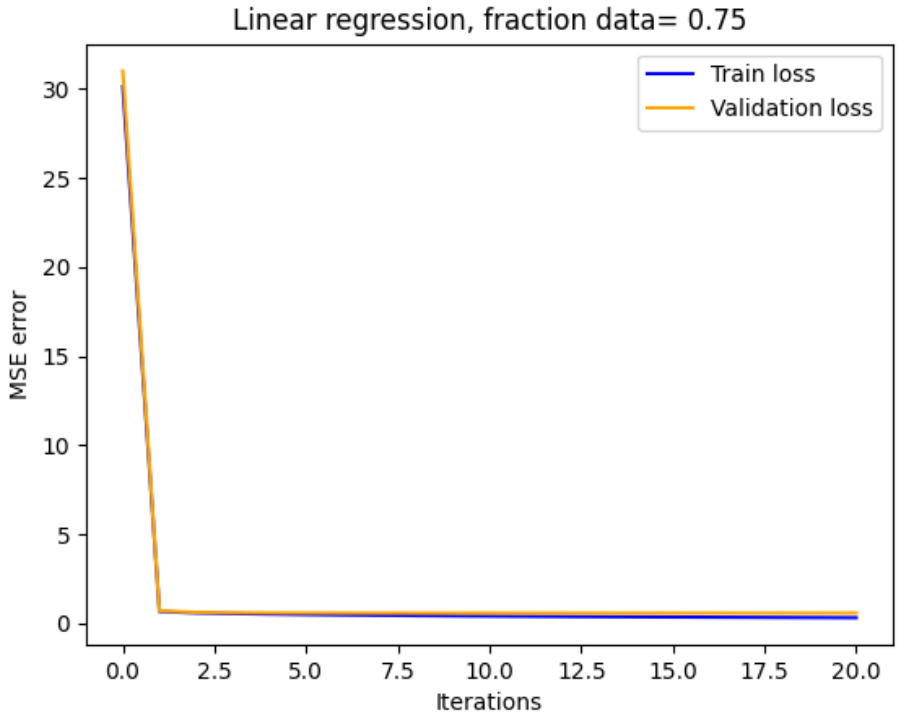
- Data fraction = 0.25, relative error threshold = 0.001, MSE Train = 0.0054, MAE Train = 0.052, MSE Val = 0.645, MAE Val = 0.671, iter = 2905



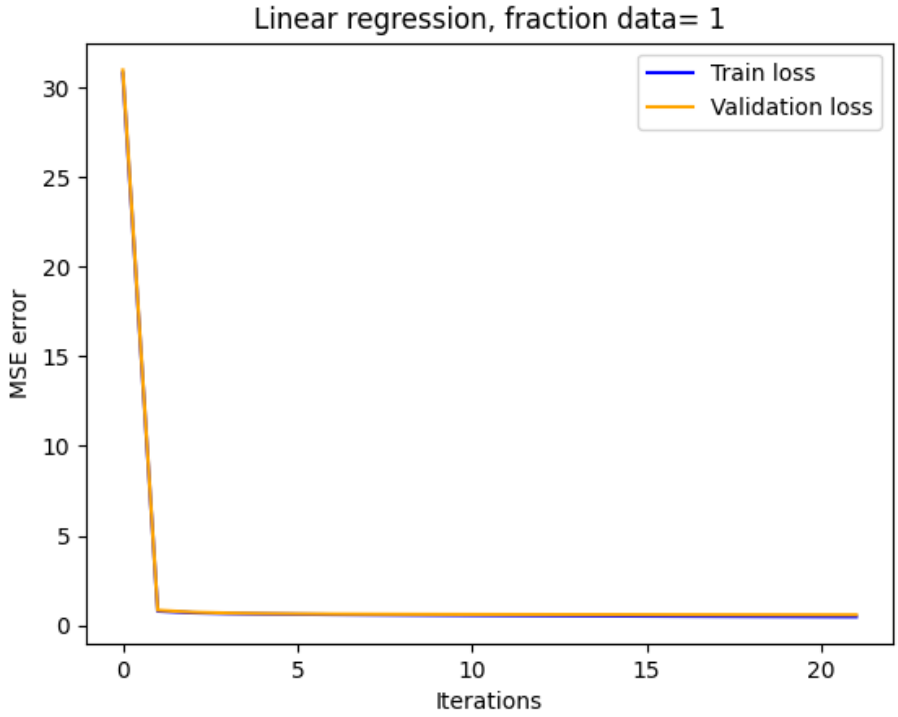
- Data fraction = 0.5, relative error threshold = 0.001, MSE Train = 0.14, MAE Train = 0.313, MSE Val = 0.599, MAE Val = 0.588, iter = 748



- Data fraction = 0.75, relative error threshold = 0.001, MSE Train = 0.317, MAE Train = 0.447, MSE Val = 0.588, MSE Val = 0.582, iter = 419



- Data fraction = 1, relative error threshold = 0.001, MSE Train = 0.463, MAE Train = 0.541, MSE Val = 0.577, iter = 237

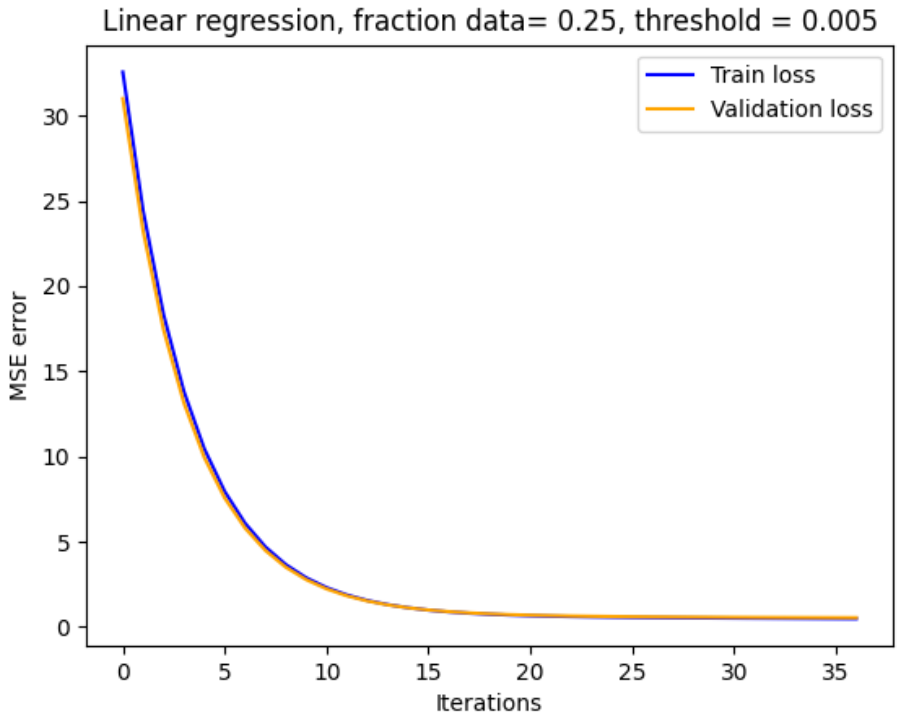


Observations - The model overfitted easily on training data when the amount of training data was low. This can be seen by the increasing training loss with increasing amount of data. However the validation loss was lower when the amount of data was high.

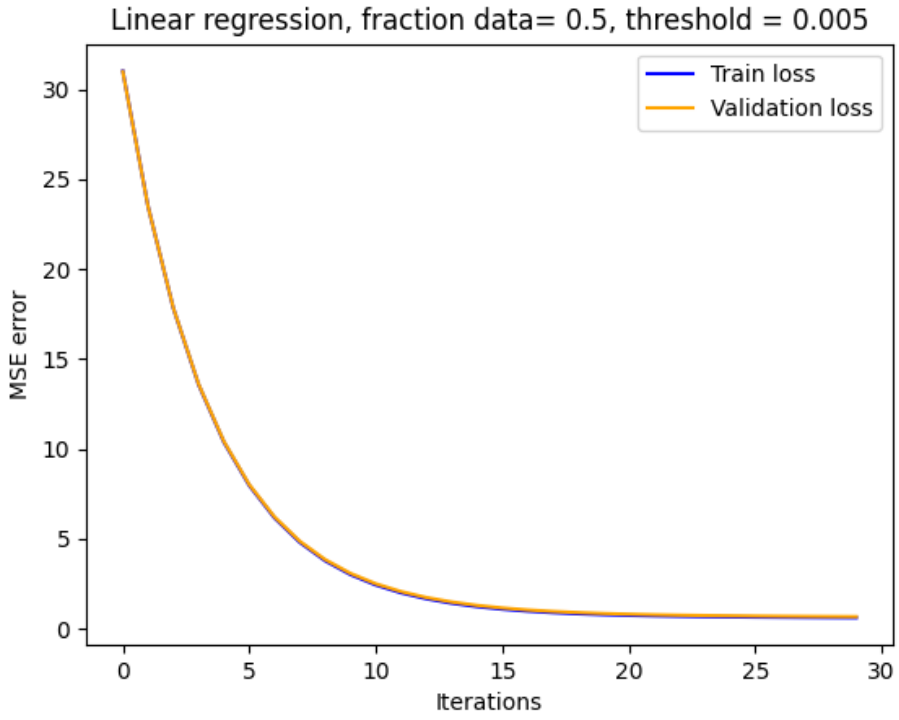
This implies that with larger amounts of data we may get higher bias but our variance would be low and we may be able to generalize better. Also the amount of iterations required to achieve same threshold of error was smaller on larger amount of data

Similar plots with lower error(0.005) threshold to decrease overfitting

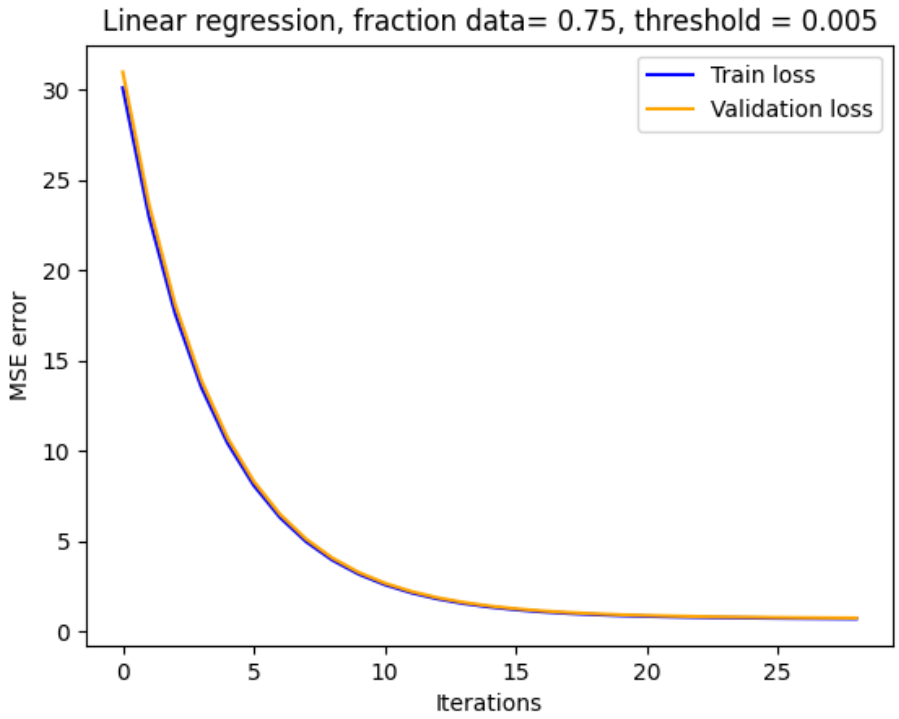
- Data fraction = 0.25, relative error threshold = 0.005



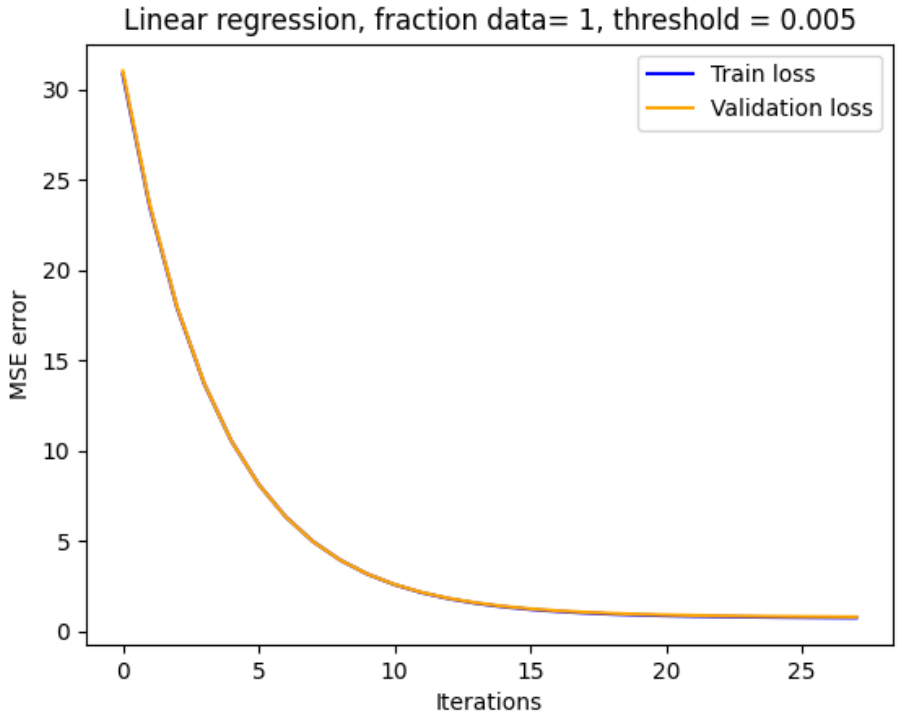
- Data fraction = 0.5, relative error threshold = 0.005



- Data fraction = 0.75, relative error threshold = 0.01



- Data fraction = 1, relative error threshold = 0.01



In this case the smallest MSE for validation data = 0.54 was obtained for data fraction = 0.25

Comparison between prediction of two halves

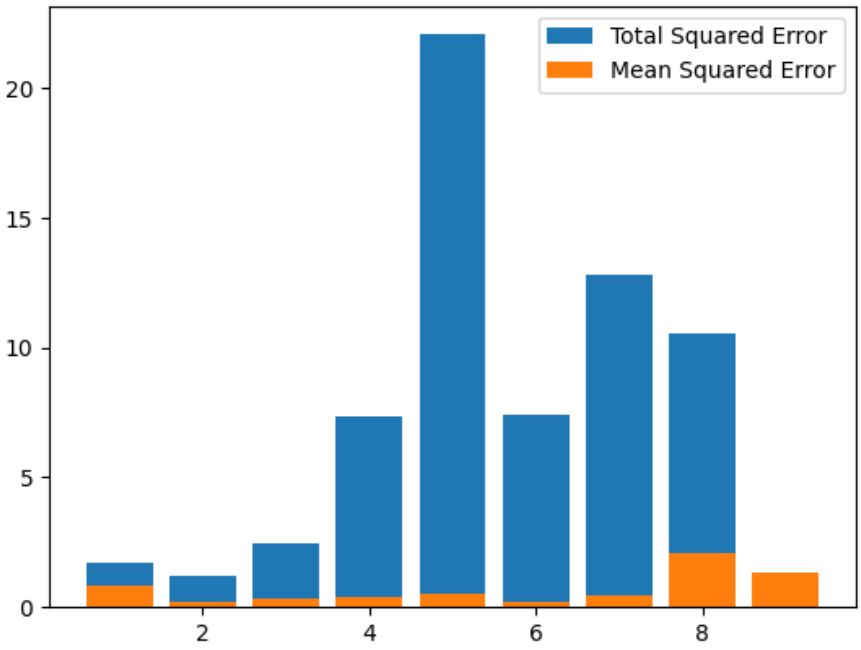
- Linear regression - The mean absolute difference between the values predicted by two models was 0.647
- Lasso regression - The mean absolute difference between the values predicted by two models

was 0.444

Observations - The model trained on different halves produced almost similar results which can be seen from the low value of mean absolute error. The mean absolute error was lower for ridge regression. This was expected as ridge regression has high bias and is less expected to overfit/depend on the dataset and will produce close/similar results even if the data is changed

Comparing MSE loss over different classes

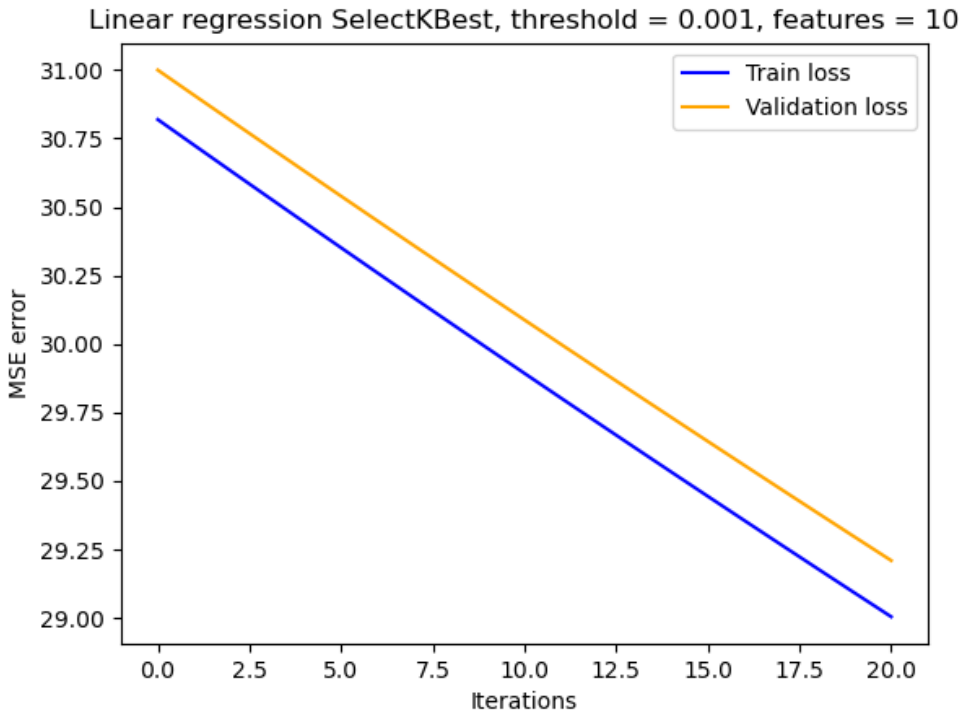
Highest Total loss was for class 5, however highest average loss was for class 8



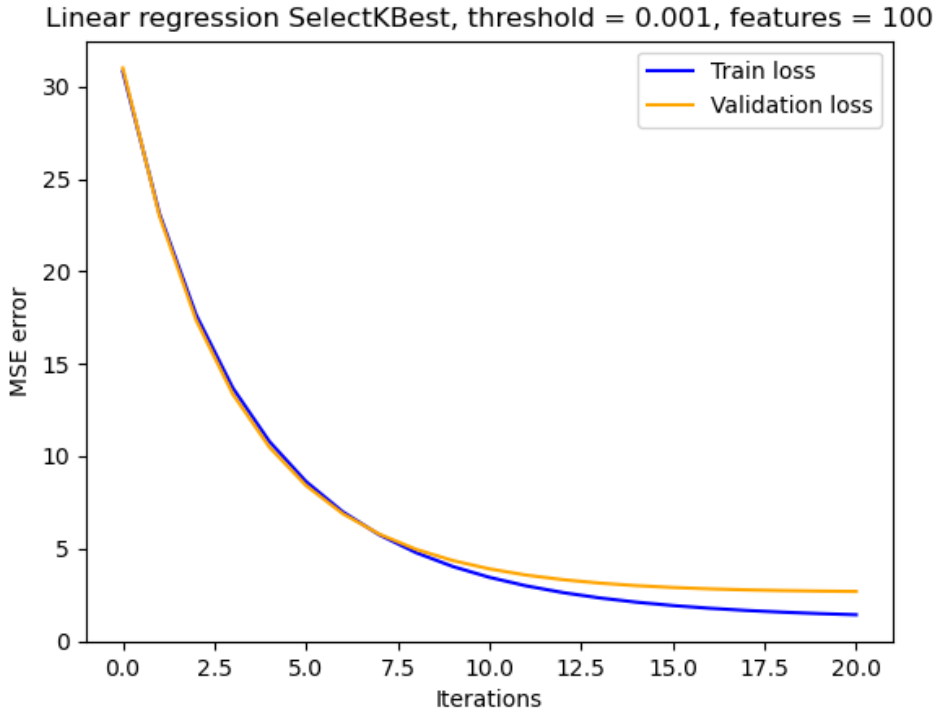
Feature selection with different number of features

I am using the SelectKBest for selecting the top features, as it gave lower MSE losses on validation data(Although SelectFromModel had better accuracies on test data, Further for SelectFromModel, the maximum number of features selected could not be increased more than 805, that is why I used SelectKBest). Note - I am using $\eta = 0.001$ for this as with higher number of features the parameters overshoot the optimal value

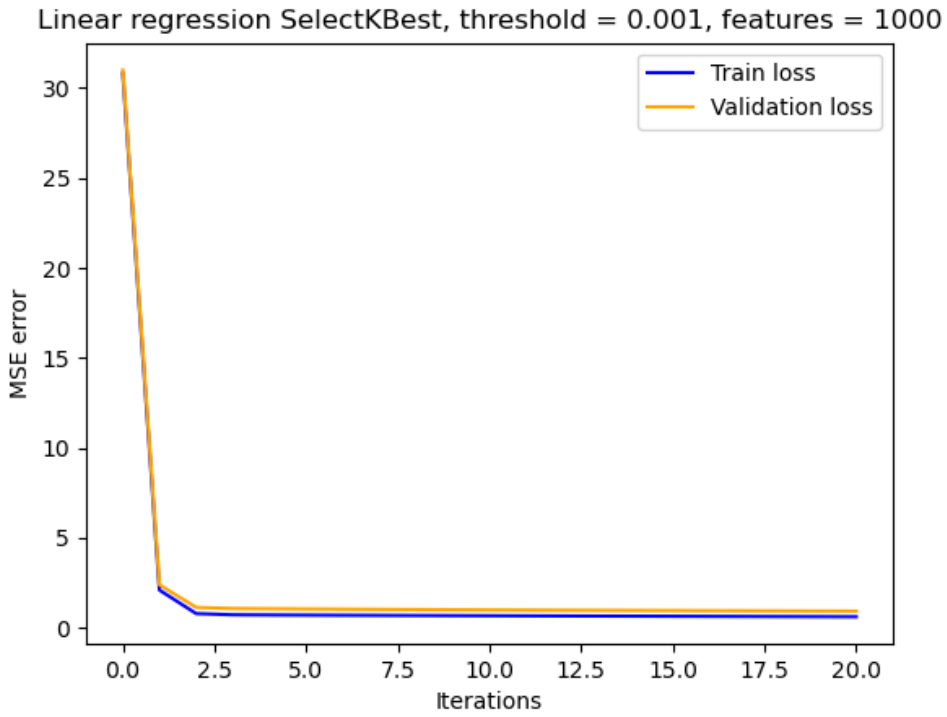
- 10 features, MSE Train = 28.946, MAE Train = 5.156, MSE Val = 29.152, MAE Val = 5.200



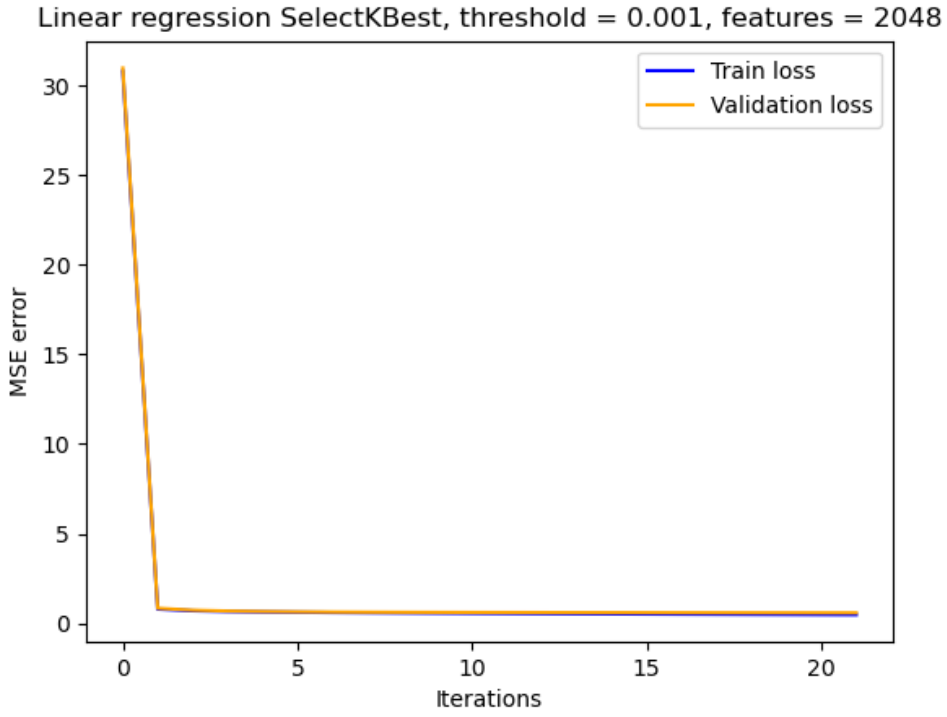
- 100 features, MSE Train = 1.398, MAE Train = 0.965, MSE Val = 2.677, MAE Val = 1.28



- 1000 features, MSE Train = 0.616, MAE Train = 0.622, MSE Val = 0.93, MAE Val = 0.594



- 2048 features, MSE Train = 0.463, MAE Train = 0.542, MSE Val = 0.574, MAE Val = 0.577

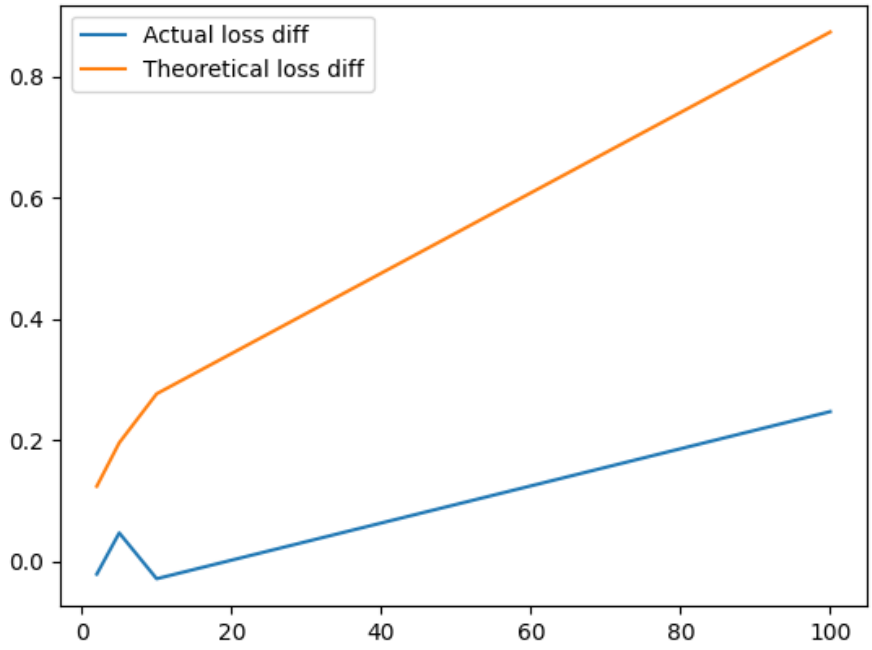


Observation - The accuracy increased on increasing the number of features for a small error threshold of 0.001 and learning rate = 0.001. This is because as the number of features are low the error threshold is achieved in very few iterations and the model may not converge to optimal value with a small step size. Hence the data with higher number of features perform better in case of smaller step size. If on the other hand step size is increased then with higher number of features overshoot happens. Also the error decreased smoothly in case of lower number of features but abruptly if number of features are high.

Generalization Analysis

The relative error threshold was taken 0.001 and the learning rate as 0.01.

- The algorithm converged in 20, 28, 30, 197 iterations for the respective dimension 2, 5, 10, 100
- The values of $E_{out} - E_{in}$ were -0.0215, 0.046, -0.029, 0.246.



The actual values of $E_{out} - E_{in}$ (blue line) are always less than $(dimension/N)^{1/2}$ (orange line, maximum theoretical value of error). Further there is a general trend of increasing of error with increasing dimension for fixed size sample

Bonus (One vs rest)

In the One-vs-Rest implementation the following is done -

- For every class C_i a binary logistic regression model is created with all elements belonging to C_i labelled as $+1$ and all other elements labelled as 0.
- This gives a set of parameters $\mathbf{w}_i$
- After getting 8 such set of parameters for every point $\mathbf{x}_i$ we estimate $P(y_i|\mathbf{x}_i, \mathbf{w}_k)$ for all classes (k) and we assign that class to the point which maximizes this value

The model was easily overfitting on training data with accuracy on training data going as high as 1.0 and on validation data about 0.33. The best validation error was achieved by taking $\eta = 0.01$ and 110 iterations. The corresponding training and validation accuracies were 0.90 and 0.43